

Path 1017

HIV & AIDS

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HIV & AIDS

- HIV- Infection with Human Immunodeficiency Virus (HIV) which affects the function of the immune system
- AIDS- Acquired Immunodeficiency Syndrome caused by the most advanced stage of HIV virus

Human Immunodeficiency Virus

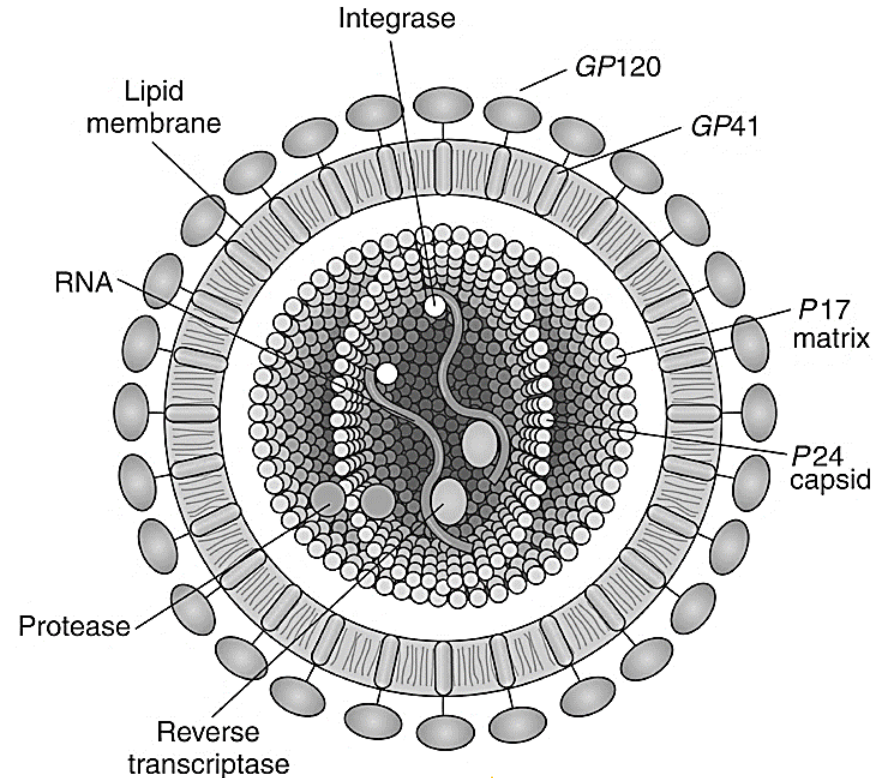
- Transmitted by body fluids
 - Sexual contact
 - Blood-to-blood contact
 - Contaminated needles
 - Transfusions
 - During pregnancy or birth
 - Breast milk

**** Patients are infectious even when asymptomatic***

What is my
occupational risk as a
nurse?

Human Immunodeficiency Virus

- Retrovirus
- HIV carries its genetic material in RNA rather than DNA
- 2 strains of HIV:
 - HIV-1
 - HIV-2
- Infects CD4 + T helper cells



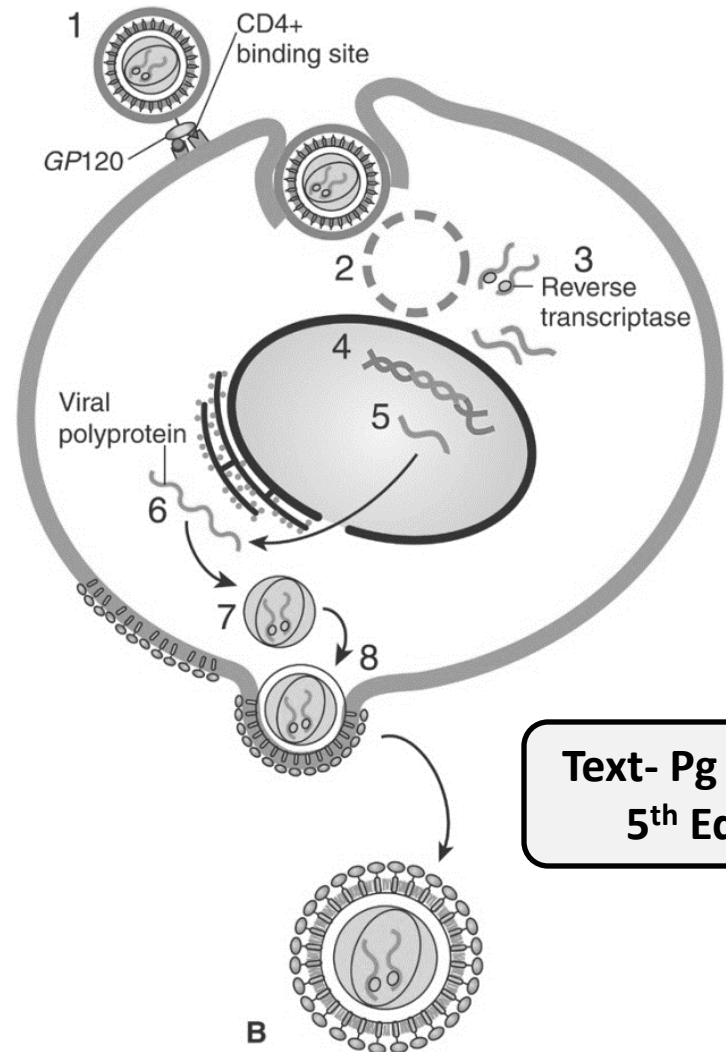
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Look at the structure of HIV
(see figure p 292 5th Ed)

HIV Infection & Replication – 8 steps

1. Attachment
2. Internalization
3. Reverse transcription
4. Integration
5. Transcription
6. Translation
7. Cleavage
8. Assembly and release



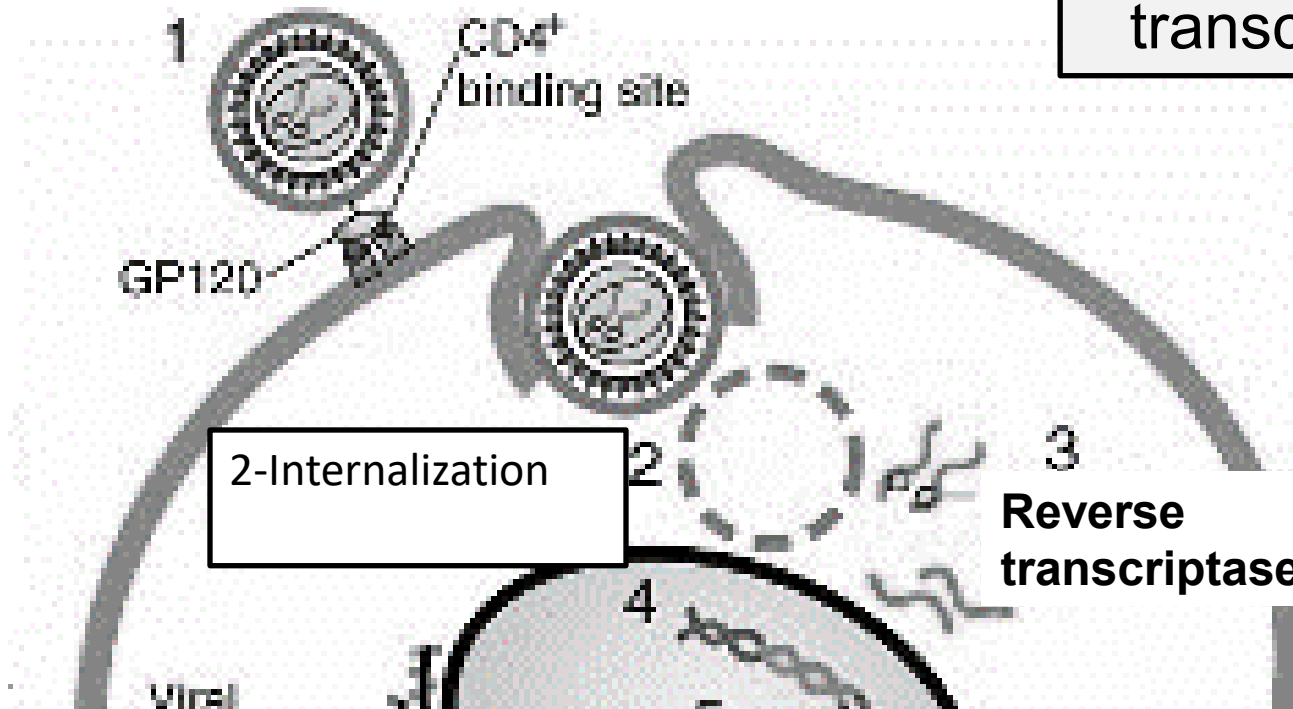
**Text- Pg 292
5th Ed**

HIV Infects a Cell (Steps 1-3)

Binds to CD4 protein receptor

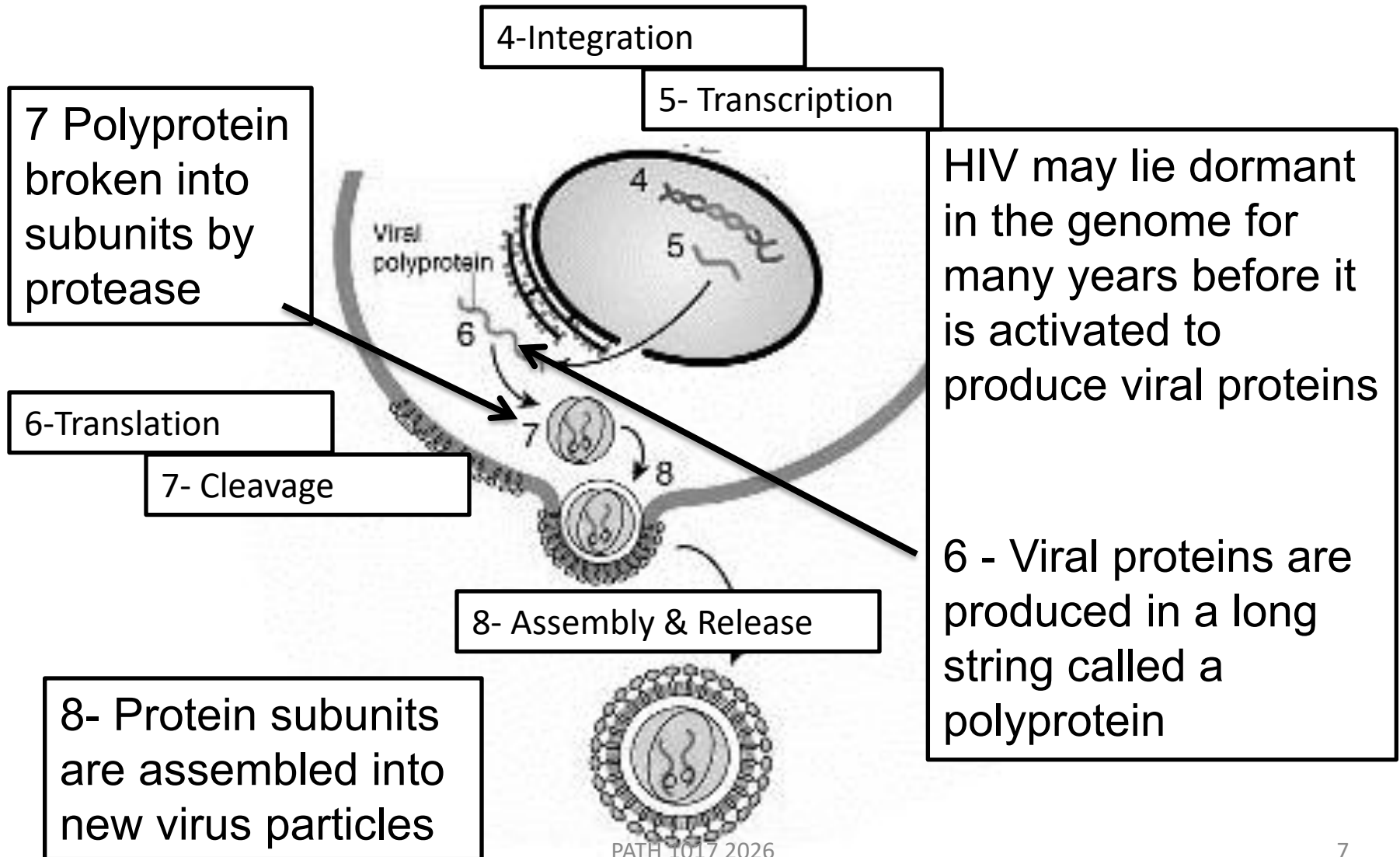
1-Attachment

- Which cells have CD4+ receptors?
- What does reverse transcriptase do?



3- Reverse
Transcription

The Infected Cell Produces New HIV (Steps 4-8)



Typical Clinical Course of HIV

Time frames?
CD4 count?
Viral load?

Window period –what is it?

1- Primary Infection Phase

- Signs of systemic infection
- **Seroconversion:** immune system response, antibodies appear (1–6m)

2- Latency Phase

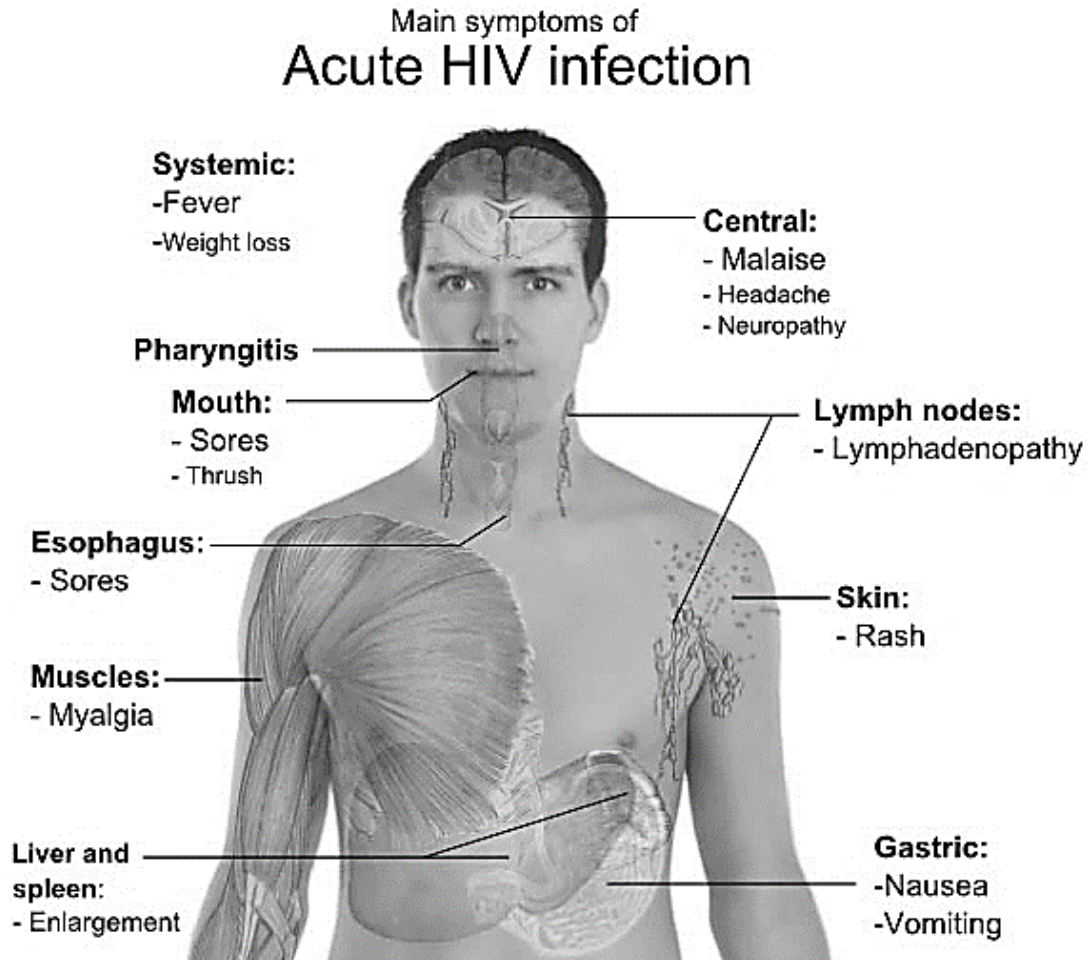
- Virus replicating
- T cell count gradually falls
- May last 10–11 years or longer

3-Overt AIDS Phase

- -T cell count <200 cells/ μ L **OR**
- AIDS-defining illness (example)
- Die in 2-3 year without treatment

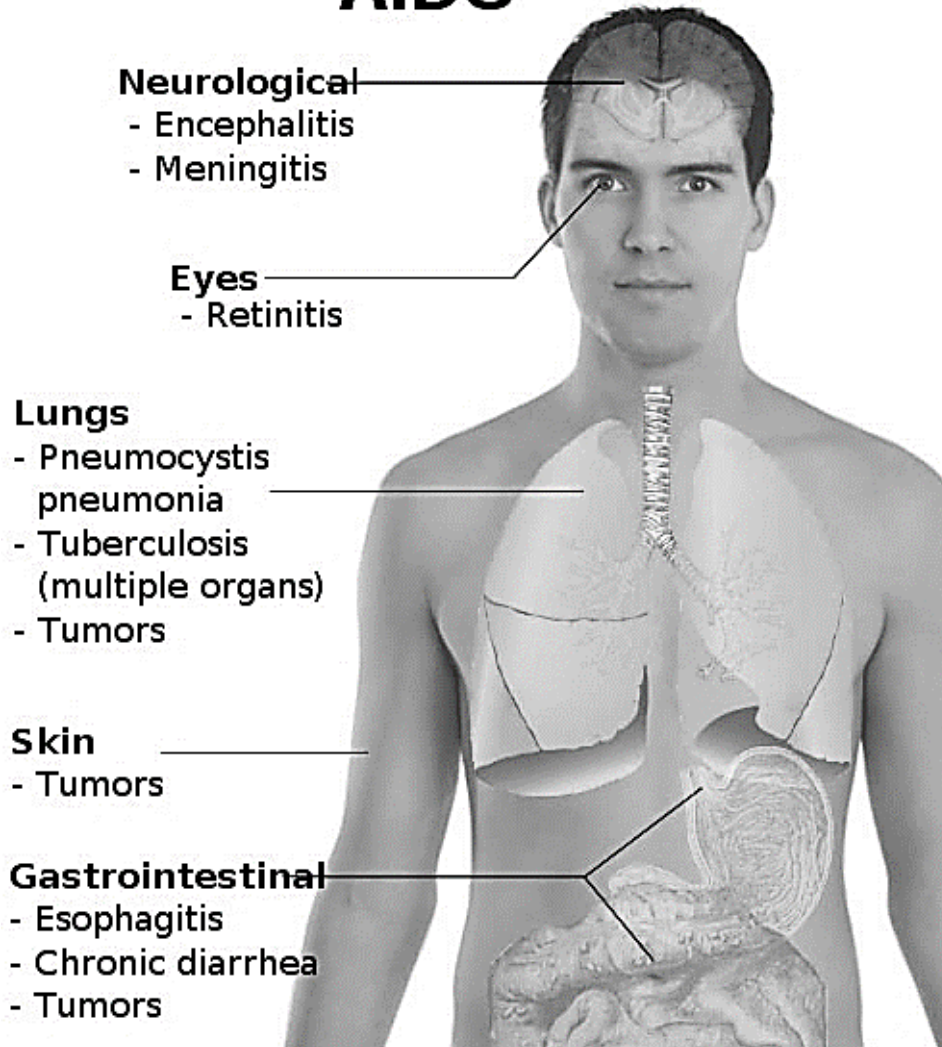
What manifestations occur in each stage of illness?

- In the primary infection stage
 - Acute mononucleosis-like symptoms
- Latent stage
 - Asymptomatic or lymphadenopathy



Overt AIDS Stage

Main symptoms of **AIDS**



Examples of AIDS-defining illnesses:

- Opportunistic infections
- Malignances
- Wasting syndrome

Diagnostic Tests

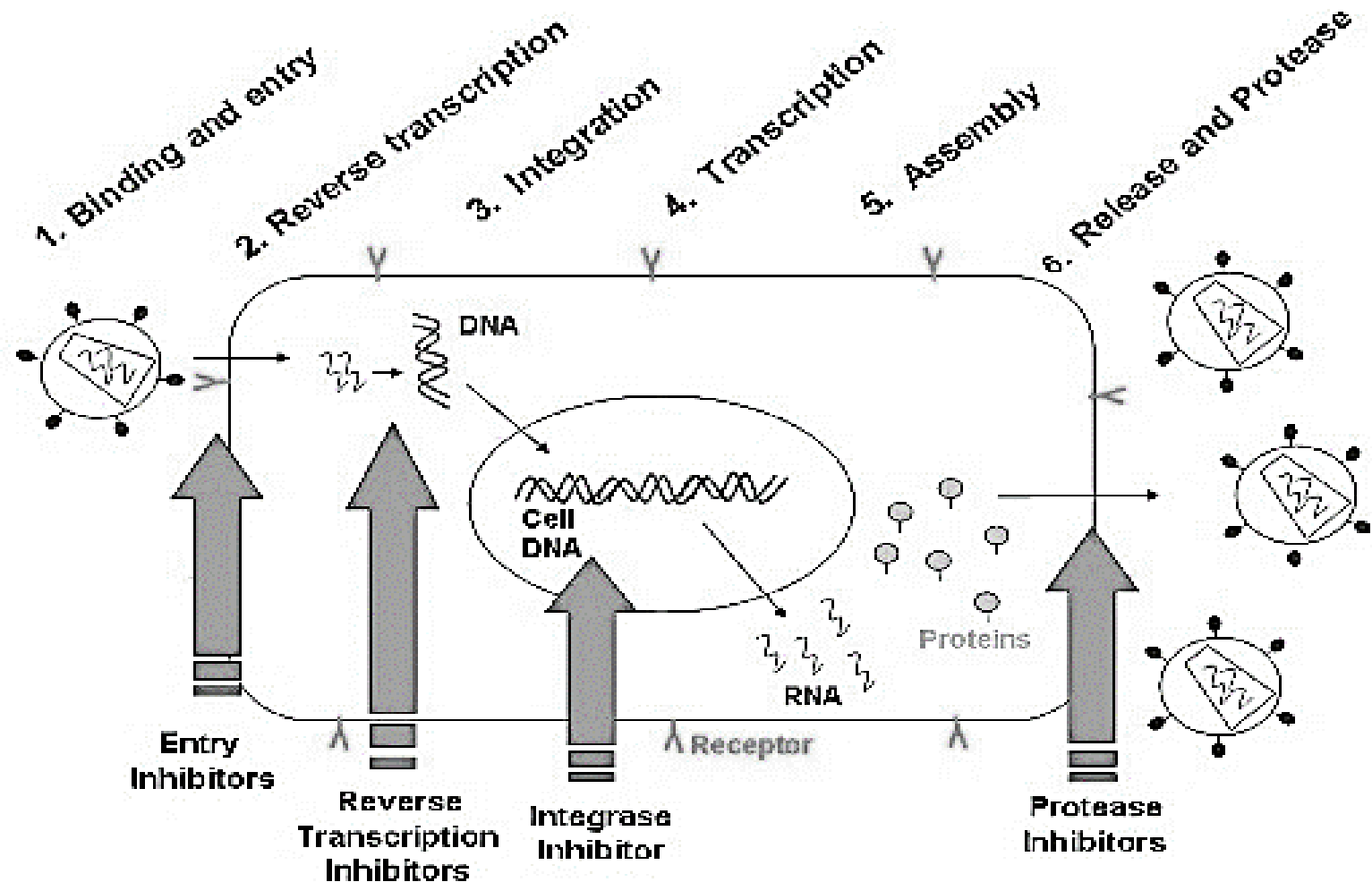
- HIV antibody tests:
 1. **ELISA** (*enzyme linked immuosorbent assay*)
 2. **Western Blot**- confirmatory test if ELISA +
- Polymerase Chain Reaction (PCR)-
 - detects HIV DNA presence rather than antibodies
- CD 4 count-
 - used to monitor impact on immune system

Is the testing the same for adults vs. infants?

Treatment

- No cure
- ART = Highly Active Antiretroviral Therapy
 - 3 drug combo (i.e. emtricitabine, tenofovir, ritonavir)
 - Goal- suppress HIV replication (↓viral load; ↑CD4 count)
 - Significant side effects
- Other:
 - Drugs & vaccines to prevent and/or treat opportunistic infections
 - PrEP- Preexposure Prophylaxis- new!

Anti-HIV Medication Target Sites



Pregnancy, Infants & Children

- Transmission of HIV to infants can be lowered by administering zidovudine to mother in pregnancy and labour and delivery and to infant when born
- HIV counselling and testing offered to all pregnant women
- Different HIV pattern in children:
 - Failure to thrive
 - CNS abnormalities
 - Developmental delays
 - Lower weight
 - Develop PCJ in earlier stage compared to adults

HIV in the community

- Stigma
 - U=U  undetectable=untransmissible
- Prevention
 - PrEP (Pre-Exposure Prophylaxis)
- PEP
 - Post-Exposure Prophylaxis